TMUA / WORKED SOLUTIONS

Set B Paper 1

Applications of Mathematical Knowledge 20 questions

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Sequences & Series

1. An infinite geometric series has non-zero first term a and common ratio r

The sum to infinity of this series is S

The sum to infinity of the squares of the terms of this series is $2S$

The sum to infinity of the cubes of the terms of this series is $\frac{36}{7}S$

What is the value of S?

A 2

B 3

C 4

D 6

E 8

F 9

G 12

Solution **Answer D**

The three series to consider are

$$a + ar + ar^2 + \cdots$$

$$a^2 + a^2r^2 + a^2r^4 + \cdots$$

$$a^3 + a^3r^3 + a^3r^6 + \cdots$$

So we see that the series of squares and series of cubes are both also geometric, with first terms a^2 and a^3 and common ratios r^2 and r^3 respectively.

Using the sum of a geometric series formula $\frac{a}{1-r}$ obtains three simultaneous equations:

$$S = \frac{a}{1-r}$$

$$2S = \frac{a^2}{1-r^2}$$

$$\frac{36}{7}S = \frac{a^3}{1-r^3}$$

First eliminate S using the original series and the series of squares:

$$\frac{a^2}{1-r^2} = \frac{2a}{1-r}$$

Clearing denominators gives

$$a^2(1-r) = 2a(1-r^2)$$

Dividing by $a \neq 0$ and factorising gives

$$a(1-r) = 2(1-r)(1+r)$$

Since $|r| < 1$, we have $1 - r \neq 0$. Hence

$$a = 2(1 + r)$$

Now eliminate S using the original series and the series of cubes:

$$\frac{a^3}{1 - r^3} = \frac{36a}{7(1 - r)}$$

Clearing denominators gives

$$7a^3(1 - r) = 36a(1 - r^3)$$

Dividing by $a \neq 0$ gives

$$7a^2(1 - r) = 36(1 - r^3)$$

Substituting $a = 2(1 + r)$ gives

$$7[2(1 + r)]^2(1 - r) = 36(1 - r^3)$$

Dividing by 4 gives

$$7(1 + r)^2(1 - r) = 9(1 - r^3)$$

Expanding and collecting terms gives a cubic:

$$2r^3 - 7r^2 + 7r - 2 = 0$$

Trying $r = 1$ gives

$$2(1)^3 - 7(1)^2 + 7(1) - 2 = 0$$

So $r - 1$ is a factor. Factorising the cubic gives

$$2r^3 - 7r^2 + 7r - 2 = (r - 1)(2r^2 - 5r + 2)$$

Factorising the quadratic gives

$$(r - 1)(2r - 1)(r - 2) = 0$$

Thus the roots of the cubic are

$$r = 1 \text{ or } r = \frac{1}{2} \text{ or } r = 2$$

The trial root $r = 1$ and the root $r = 2$ both fail $|r| < 1$. Therefore

$$r = \frac{1}{2}$$

Back-substituting gives

$$a = 2\left(1 + \frac{1}{2}\right) = 3$$

and then

$$S = \frac{3}{1 - \frac{1}{2}} = 6$$

The correct option is **D**.

Alternative Solution:

Recall the difference-of-two-squares identity

$$a^2 - b^2 = (a - b)(a + b)$$

The lesser known difference-of-two-cubes identity is

$$a^3 - b^3 = (a - b)(a^2 + ab + b^2)$$

With $a = 1$ and $b = r$ this identity reads

$$1 - r^3 = (1 - r)(1 + r + r^2)$$

We can prove this identity by simply expanding the right hand side

$$\begin{aligned}(1 - r)(1 + r + r^2) &= (1 + r + r^2) - (r + r^2 + r^3) \\ &= 1 - r^3\end{aligned}$$

And, in the general case:

$$\begin{aligned}(a - b)(a^2 + ab + b^2) &= (a^3 + a^2b + ab^2) - (a^2b + ab^2 + b^3) \\ &= a^3 - b^3\end{aligned}$$

Note that $1 - r^3$ is the denominator of the sum of the cubes series. The presence of the $(1 - r)$ factor in the difference of two cubes factorisation allows us to leverage a cancellation and avoid solving a cubic.

For the non-zero case, $a \neq 0$ and $|r| < 1$. The three geometric-series sums give

$$S = \frac{a}{1 - r}$$

$$2S = \frac{a^2}{1 - r^2}$$

$$\frac{36}{7}S = \frac{a^3}{1 - r^3}$$

Using $1 - r^2 = (1 - r)(1 + r)$ in the first two equations gives

$$\frac{a^2}{(1 - r)(1 + r)} = \frac{2a}{1 - r}$$

Cancelling a and $1 - r$, both non-zero, gives

$$\frac{a}{1 + r} = 2$$

Hence

$$a = 2(1 + r)$$

To avoid a cubic, factorise $1 - r^3$ before using the third equation. Using difference-of-two-cubes

$$1 - r^3 = (1 - r)(1 + r + r^2)$$

Equating the first and third sums now gives

$$\frac{a^3}{(1-r)(1+r+r^2)} = \frac{36a}{7(1-r)}$$

Cancelling a and $1-r$ gives

$$\frac{a^2}{1+r+r^2} = \frac{36}{7}$$

Substituting $a = 2(1+r)$ gives

$$\frac{[2(1+r)]^2}{1+r+r^2} = \frac{36}{7}$$

Rearranging gives

$$7(1+r)^2 = 9(1+r+r^2)$$

Expanding and collecting terms gives

$$2r^2 - 5r + 2 = 0$$

Factorising gives

$$(2r-1)(r-2) = 0$$

Thus

$$r = \frac{1}{2} \text{ or } r = 2$$

Since $|r| < 1$, we take

$$r = \frac{1}{2}$$

Therefore

$$a = 2\left(1 + \frac{1}{2}\right) = 3$$

and

$$S = \frac{3}{1 - \frac{1}{2}} = 6$$

So the correct option is again **D**.

Algebra

2. A straight line with gradient m passes through the point $(1, 4)$

The line intersects the curve $y = x^2$ at two distinct points with x-coordinates x_1 and x_2

What is the minimum possible value of $x_1^2 + x_2^2$ as m varies over all real numbers?

A 1

B 4

C 7

D 8

E 12

F 16

Solution **Answer C**

The line through $(1, 4)$ with gradient m has equation

$$y - 4 = m(x - 1)$$

At an intersection with $y = x^2$, this becomes

$$x^2 - mx + m - 4 = 0$$

Recall that for a quadratic $ax^2 + bx + c = 0$ the sum of the roots is given by $-b/a$ and the product of the roots is given by c/a for $a \neq 0$.

In this case, we get

$$x_1 + x_2 = m \qquad \text{and} \qquad x_1 x_2 = m - 4$$

Recalling the identity

$$x_1^2 + x_2^2 = (x_1 + x_2)^2 - 2x_1 x_2$$

Substituting gives

$$x_1^2 + x_2^2 = m^2 - 2(m - 4)$$

Completing the square,

$$\begin{aligned} x_1^2 + x_2^2 &= m^2 - 2m + 8 \\ &= (m - 1)^2 + 7 \end{aligned}$$

The squared term is at least 0, so the minimum is 7, attained when $m = 1$.

This value of m is allowed: the intersection equation then has discriminant

$$(-1)^2 - 4(1)(-3) = 13 > 0$$

so the two points are distinct.

The minimisation can also be done by differentiation.

Differentiating $m^2 - 2m + 8$ gives $2m - 2$, which is zero at $m = 1$; the quadratic opens upwards, so this is its minimum. Substituting $m = 1$ into $x_1^2 + x_2^2$ obtains the minimum of 7.

In either case we get that the correct option is **C**.

Alternative Solution:

The intersection equation is

$$x^2 - mx + m - 4 = 0$$

Its discriminant is

$$\begin{aligned} m^2 - 4(m-4) &= m^2 - 4m + 16 \\ &= (m-2)^2 + 12 > 0 \end{aligned}$$

Thus the quadratic formula gives two distinct roots for every real m:

$$x_1 = \frac{m - \sqrt{m^2 - 4m + 16}}{2}$$

and

$$x_2 = \frac{m + \sqrt{m^2 - 4m + 16}}{2}$$

Squaring each expression gives

$$x_1^2 = \frac{m^2 - 2m\sqrt{m^2 - 4m + 16} + m^2 - 4m + 16}{4}$$

and

$$x_2^2 = \frac{m^2 + 2m\sqrt{m^2 - 4m + 16} + m^2 - 4m + 16}{4}$$

The terms containing the square root cancel when we add:

$$x_1^2 + x_2^2 = \frac{2m^2 + 2(m^2 - 4m + 16)}{4}$$

Simplifying and completing the square,

$$\begin{aligned} x_1^2 + x_2^2 &= m^2 - 2m + 8 \\ &= (m-1)^2 + 7 \end{aligned}$$

Its minimum is 7, attained at $m = 1$, so the correct option is **C**.

Finding the roots separately works, but sums and products of roots reach the expression to minimise with less algebra.

Trigonometry

3. How many distinct real values of x satisfy the equation

$$\cos(2x) = 1 - x$$

in the range $0 \leq x \leq \pi$?

A 0

B 1

C 2

D 3

E 4

F 5

G 6

Solution **Answer D**

Sketch $y = \cos(2x)$ and $y = 1 - x$ on the same axes. The cosine curve starts at 1, falls to -1 at $x = \pi/2$, then rises to 1 at $x = \pi$. The straight line starts at 1 and has gradient -1.

At $x = 0$, the graphs meet:

$$\cos(0) = 1 = 1 - 0$$

At $x = \pi/4$, the cosine curve is below the line, since $\pi/4 < 1$:

$$\cos\left(2 \cdot \frac{\pi}{4}\right) = 0 < 1 - \frac{\pi}{4}$$

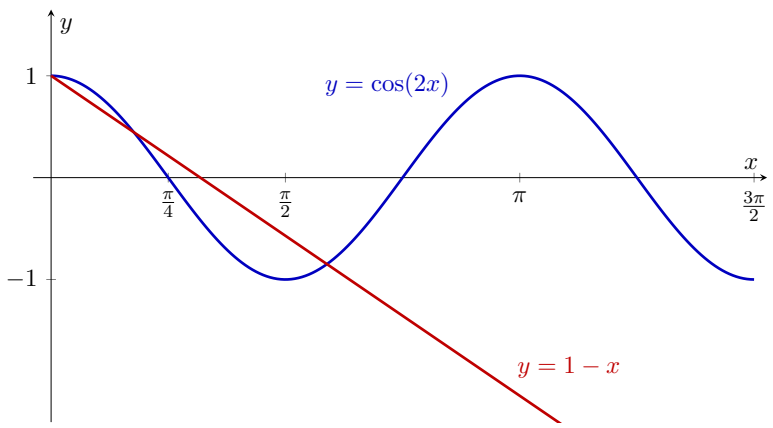
It is still below the line at $x = \pi/2$, since $\pi/2 < 2$:

$$\cos\left(2 \cdot \frac{\pi}{2}\right) = -1 < 1 - \frac{\pi}{2}$$

By $x = \pi$, it is above the line:

$$\cos(2\pi) = 1 > 1 - \pi$$

These values and the shape of the cosine curve give the following sketch.



Our sketch shows us three intersection points.

Therefore there are three real solution to the original equation and the correct option is then **D**.

Integration

4. Use the trapezium rule with 4 equal strips to estimate

$$\int_1^5 \ln x \, \mathrm{d}x$$

A $\frac{1}{2}\ln(120)$

B $\frac{1}{2}\ln(180)$

C $\frac{1}{2}\ln(240)$

D $\frac{1}{2}\ln(720)$

E $\frac{1}{2}\ln(2880)$

F $\frac{1}{2}\ln(14400)$

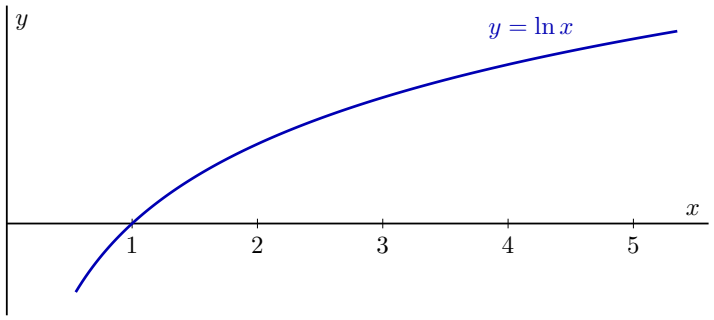
Solution **Answer E**

For n equal strips, the trapezium rule is

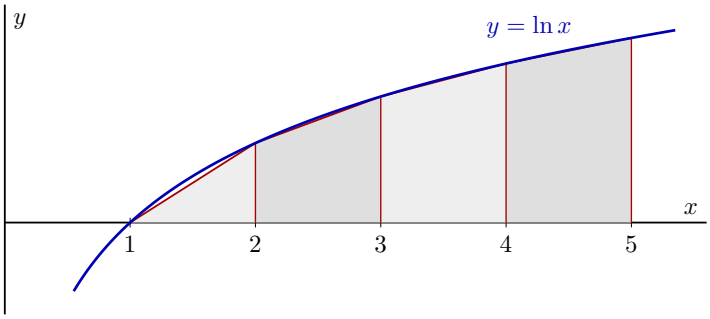
$$\int_{x_0}^{x_n} y \, \mathrm{d}x \approx \frac{h}{2}\left[y_0 + y_n + 2(y_1 + \cdots + y_{n-1})\right]$$

Here h is the width of each strip, and y_i is the height of the curve at the ith division point. The end heights y_0 and y_n are counted once; each height between them is shared by two trapezia and counted twice.

First, sketch the curve over the interval being estimated.



On the same curve, draw straight edges between the heights at $x = 1, 2, 3, 4, 5$. These form the four strips used by the rule.



The strips have width

$$h = \frac{5-1}{4} = 1$$

The five heights are

$$\begin{aligned} y_0 &= \ln 1 = 0 \\ y_1 &= \ln 2 \\ y_2 &= \ln 3 \\ y_3 &= \ln 4 \\ y_4 &= \ln 5 \end{aligned}$$

Substituting into the rule gives

$$\int_1^5 \ln x \, \mathrm{d}x \approx \frac{1}{2}\big[0 + \ln 5 + 2(\ln 2 + \ln 3 + \ln 4)\big]$$

Use the logarithm laws one step at a time:

$$\begin{aligned} \int_1^5 \ln x \, \mathrm{d}x &\approx \frac{1}{2}\big[\ln 5 + 2(\ln 6 + \ln 4)\big] \\ &= \frac{1}{2}\big[\ln 5 + 2 \ln 24\big] \\ &= \frac{1}{2}\big[\ln 5 + \ln(24^2)\big] \\ &= \frac{1}{2}\big[\ln 5 + \ln 576\big] \\ &= \frac{1}{2}\ln(5 \times 576) \\ &= \frac{1}{2}\ln 2880 \end{aligned}$$

The correct option is **E**.

Coordinate Geometry

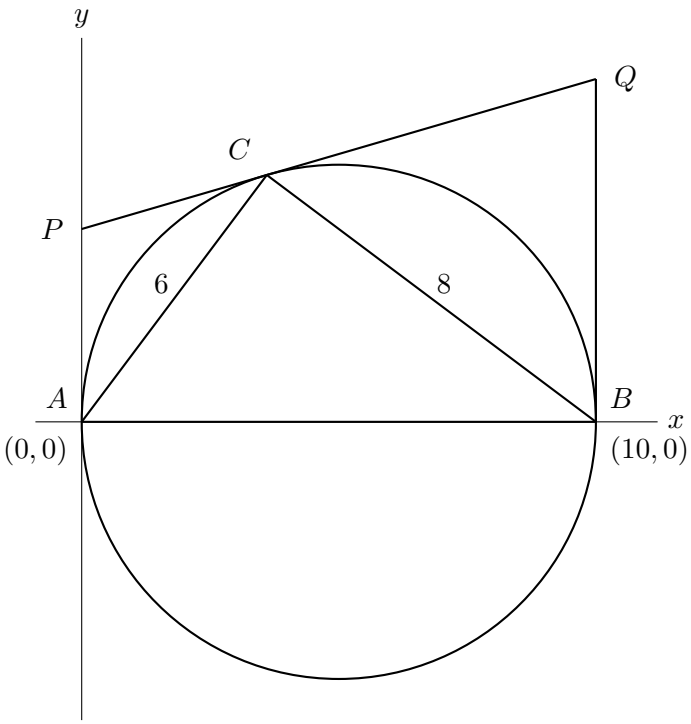
5. The points A and B have coordinates $(0, 0)$ and $(10, 0)$ respectively.

A circle has diameter AB.

The point C lies on the circle above the x-axis, with $AC = 6$ and $BC = 8$.

The tangent to the circle at C meets the lines $x = 0$ and $x = 10$ at P and Q respectively.

What is the length PQ?



A $\frac{35}{12}$

B $\frac{15}{4}$

C $\frac{20}{3}$

D 10

E $\frac{125}{12}$

F $\frac{125}{6}$

Solution **Answer E**

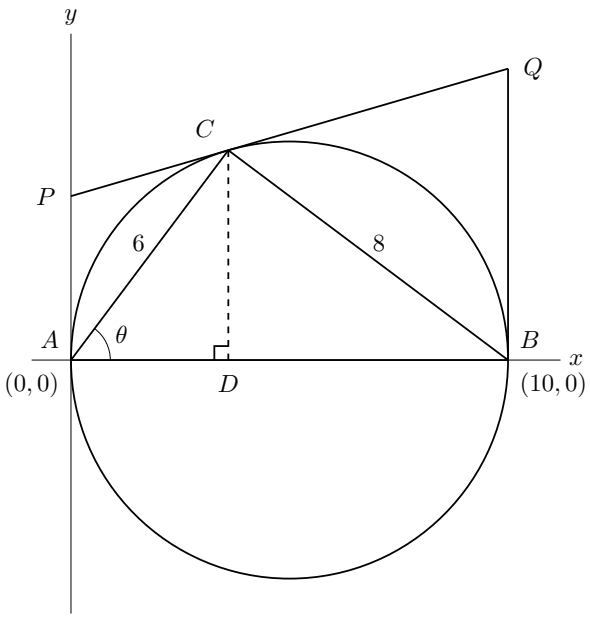
Because AB is a diameter, the angle in the semicircle is a right angle:

$$\angle ACB = 90^\circ$$

Let O be the centre of the circle, and let D be the foot of the perpendicular from C to AB.

Write $\theta = \angle BAC$.

The following sketch shows the resulting geometry:



In right-angled triangle BAC, $AB = 10$, so

$$\sin\theta = \frac{BC}{AB} = \frac{8}{10} = \frac{4}{5}$$

Similarly,

$$\cos\theta = \frac{AC}{AB} = \frac{6}{10} = \frac{3}{5}$$

Now use right-angled triangle ACD. Its horizontal side is

$$AD = AC\cos\theta = 6\left(\frac{3}{5}\right) = \frac{18}{5}$$

Its vertical side is

$$CD = AC\sin\theta = 6\left(\frac{4}{5}\right) = \frac{24}{5}$$

Since $A = (0, 0)$ and C is above the x-axis, these are the coordinates of C:

$$C = \left(\frac{18}{5}, \frac{24}{5}\right)$$

The centre is the midpoint of AB, so

$$O = (5, 0)$$

Hence the gradient of the radius OC is

$$\begin{aligned}\text{gradient of } OC &= \frac{\frac{24}{5} - 0}{\frac{18}{5} - 5} \\ &= \frac{\frac{24}{5}}{-\frac{7}{5}} \\ &= -\frac{24}{7}\end{aligned}$$

The tangent PQ is perpendicular to OC, so its gradient is the negative reciprocal:

$$\text{gradient of } PQ = -\frac{1}{-\frac{24}{7}} = \frac{7}{24}$$

Draw a horizontal line from P to meet the vertical line BQ at R. The horizontal distance from $x = 0$ to $x = 10$ is 10, so

$$PR = 10$$

The gradient of PQ is its vertical rise divided by its horizontal run. Thus

$$\frac{RQ}{PR} = \frac{7}{24}$$

Substituting $PR = 10$ gives

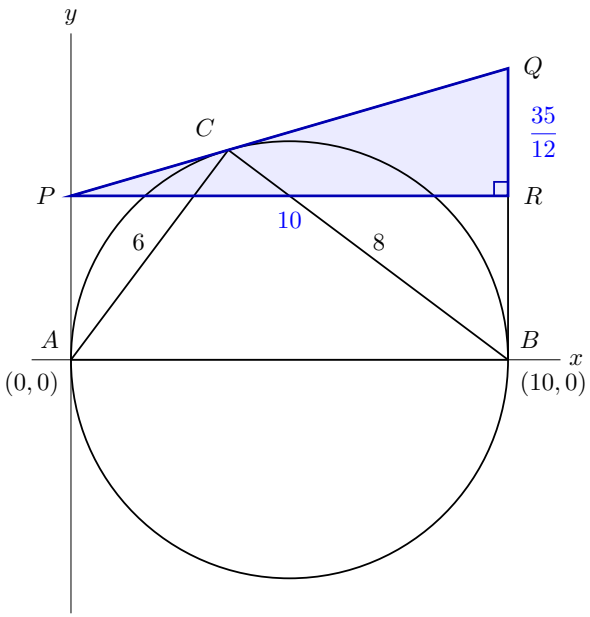
$$\frac{RQ}{10} = \frac{7}{24}$$

Therefore

$$RQ = \frac{70}{24}$$

Simplifying,

$$RQ = \frac{35}{12}$$



Triangle PRQ is right-angled at R. By Pythagoras,

$$PQ = \sqrt{PR^2 + RQ^2}$$

Substituting the two lengths,

$$PQ = \sqrt{10^2 + \left(\frac{35}{12}\right)^2}$$

To simplify by hand, write both lengths as multiples of $\frac{5}{12}$:

$$PQ = \sqrt{\left(\frac{5}{12} \times 24\right)^2 + \left(\frac{5}{12} \times 7\right)^2}$$

Factor out the common square before evaluating the remaining squares:

$$\begin{aligned} PQ &= \sqrt{\left(\frac{5}{12}\right)^2 (24^2 + 7^2)} \\ &= \frac{5}{12}\sqrt{24^2 + 7^2} \\ &= \frac{5}{12}\sqrt{576 + 49} \\ &= \frac{5}{12}\sqrt{625} \\ &= \frac{5 \times 25}{12} \end{aligned}$$

Therefore

$$PQ = \frac{125}{12}$$

The correct option is **E**.

Integration

6. Let k be a positive real constant. The function $f(x)$ is defined for $x > 0$ by

$$f(x) = \int_1^x \frac{t-k}{\sqrt{t}}\,\mathrm{d}t$$

What is the complete set of values of k for which the equation $f(x) = 0$ has exactly two distinct real solutions for $x > 0$?

A $k > 1$

B $k > \frac{1}{3}$

C $k > \frac{1}{3}$ and $k \neq 1$

D $\frac{1}{3} < k < 1$

E $k > 0$ and $k \neq 1$

F $k \geq \frac{1}{4}$ and $k \neq 1$

Solution **Answer C**

First evaluate the integral:

$$\begin{aligned} f(x) &= \int_1^x \left(t^{1/2} - kt^{-1/2}\right)\,\mathrm{d}t \\ &= \left[\frac{2}{3}t^{3/2} - 2kt^{\frac{1}{2}}\right]_1^x \\ &= \frac{2}{3}(x^{3/2} - 1) - 2k(\sqrt{x} - 1) \end{aligned}$$

Set $y = \sqrt{x}$. Since $x > 0$, we require $y > 0$, and each positive value of y corresponds to exactly one value of x. The equation $f(x) = 0$ becomes

$$y^3 - 3ky + 3k - 1 = 0$$

We already know that $x = 1$ is a solution, because the integral from 1 to 1 is zero. Thus $y = 1$ is a root of this cubic, so $y - 1$ is a factor.

Dividing the cubic by $y - 1$ gives

$$(y-1)(y^2 + y + 1 - 3k) = 0$$

The root $y = 1$ is always present. To obtain exactly two distinct positive roots overall, the quadratic must have either one positive root different from 1, or two distinct positive roots, one of which is 1.

Using the quadratic formula, the roots of

$$y^2 + y + 1 - 3k = 0$$

are

$$y = \frac{-1 \pm \sqrt{1 - 4(1-3k)}}{2} = \frac{-1 \pm \sqrt{12k-3}}{2}$$

When these roots are real, the root

$$\frac{-1-\sqrt{12k-3}}{2}$$

is always negative. Thus the quadratic cannot have two positive roots. We need its other root to be positive and different from 1.

For this root to be positive,

$$\frac{-1+\sqrt{12k-3}}{2} > 0$$

Hence

$$\sqrt{12k-3} > 1$$

Squaring and rearranging gives

$$12k-3>1 \quad \Longrightarrow \quad k > \frac{1}{3}$$

We must also exclude the case where this root equals 1:

$$\frac{-1+\sqrt{12k-3}}{2} = 1 \quad \Longrightarrow \quad \sqrt{12k-3} = 3 \quad \Longrightarrow \quad k = 1$$

Therefore the equation has exactly two distinct solutions for $x > 0$ precisely when

$$k > \frac{1}{3} \quad \text{and} \quad k \neq 1$$

The correct option is **C**.

As a shortcut, the quadratic equation can be written as

$$y^2 + y = 3k - 1$$

For $y > 0$, the left-hand side increases from 0 without bound, so there is exactly one positive root precisely when $k > \frac{1}{3}$. Substituting $y = 1$ then identifies $k = 1$ as the value to exclude.

Alternative Solution: (Graph Sketching)

The Fundamental Theorem of Calculus, which is on the TMUA specification, tells us that differentiating an integral with respect to its upper limit gives the integrand evaluated at that limit:

$$\frac{\mathrm{d}}{\mathrm{d}x}\left(\int_a^x g(t)\,\mathrm{d}t\right) = g(x)$$

Applying this here gives

$$f'(x) = \frac{x-k}{\sqrt{x}}$$

Since $\sqrt{x} > 0$, the graph decreases for $0 < x < k$ and increases for $x > k$. It therefore has a minimum at $x = k$, and can cross the axis at most once on each side of this minimum.

We also know that

$$f(1) = \int_1^1 \frac{t-k}{\sqrt{t}}\,\mathrm{d}t = 0$$

To determine whether there is another zero, we need to know how the graph behaves near $x = 0$ and for large positive x. Evaluating the integral gives

$$f(x) = \frac{2}{3}(x^{3/2}-1) - 2k(\sqrt{x}-1)$$

As x approaches 0 from above, both $x^{3/2}$ and $\sqrt{x}$ approach 0, so $f(x)$ approaches

$$2k - \frac{2}{3}$$

For large positive x, the $x^{3/2}$ term dominates the $\sqrt{x}$ term, so $f(x)$ increases without bound.

First suppose $0 < k < 1$. The minimum lies to the left of $x = 1$ and is below the axis, since the graph increases from that minimum to $f(1) = 0$. There is a second crossing before the minimum precisely when the graph starts above the axis near $x = 0$:

$$2k - \frac{2}{3} > 0$$

Thus, within this case, there are two distinct solutions precisely when

$$\frac{1}{3} < k < 1$$

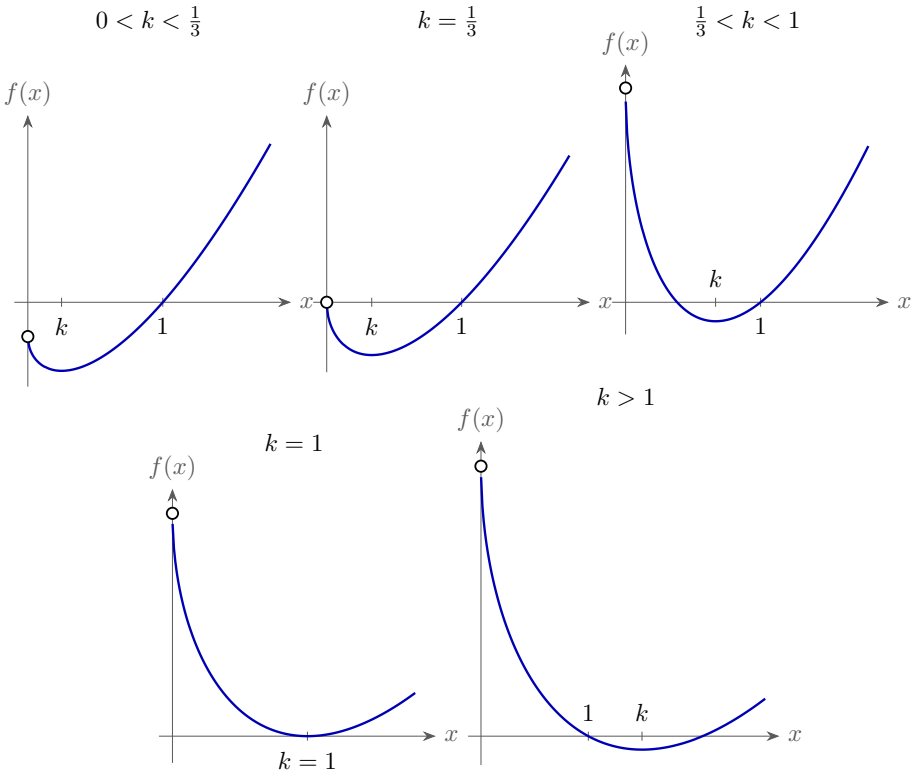
If $k = \frac{1}{3}$, the graph approaches height 0 as x approaches 0, but $x = 0$ is excluded. The graph decreases below the axis before reaching its minimum, so there is no additional solution.

If $0 < k < \frac{1}{3}$, the graph is already below the axis near $x = 0$ and decreases further before reaching its minimum. Again, there is no additional solution.

If $k = 1$, the minimum occurs at $x = 1$ and has value 0. The graph touches the axis there, giving only one solution.

If $k > 1$, the graph decreases from $f(1) = 0$ to a negative minimum at $x = k$. It then increases without bound, so it crosses the axis exactly once more after $x = k$.

The sketches below summarise these five cases. The first two have just one crossing, while the third has two. When $k = 1$, the graph only touches the axis; when $k > 1$, it crosses twice. Each open circle marks the height approached at $x = 0$, which is excluded from the domain.



Combining the two cases with exactly two distinct solutions gives

$$\frac{1}{3} < k < 1 \quad \text{or} \quad k > 1$$

Equivalently,

$$k > \frac{1}{3} \quad \text{and} \quad k \neq 1$$

The correct option is **C**.

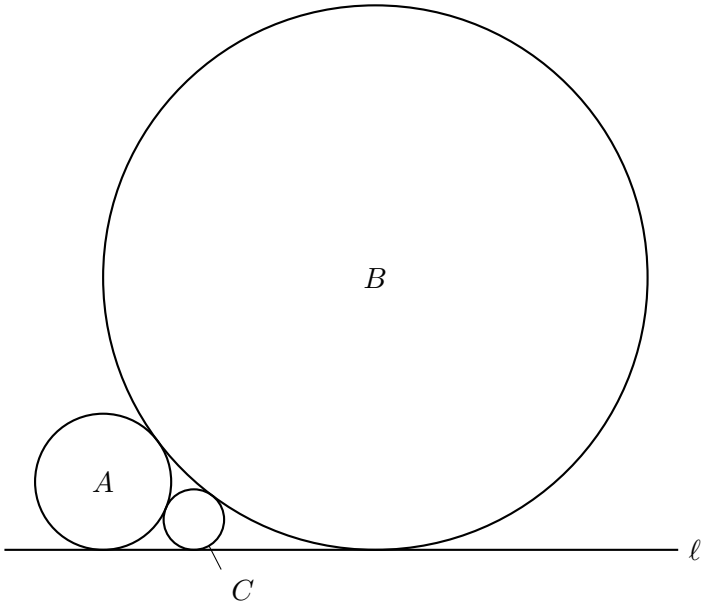
Geometry

7. A straight line ℓ is drawn, and three circles lie on the same side of ℓ. Each circle touches ℓ at exactly one point.

Circle A has radius 1 and circle B has radius 4, and these two circles touch each other.

Circle C lies in the space between circle A, circle B and the line ℓ, touching circle A, circle B and ℓ.

What is the radius of circle C?



A $\frac{2}{9}$

B $\frac{4}{9}$

C $\frac{1}{2}$

D $\frac{16}{25}$

E $\frac{2}{3}$

F $\frac{4}{5}$

G $\frac{8}{9}$

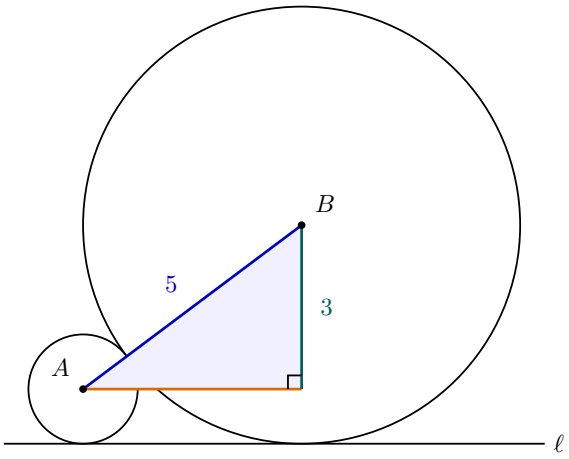
Solution **Answer B**

Let r be the radius of circle C. For two circles that touch externally, the distance between their centres is the sum of their radii.

Because each circle touches ℓ, its centre is one radius above the line.

First consider circles A and B. Their centres are $1 + 4 = 5$ apart, and the difference in their heights is $4 - 1 = 3$.

These are the hypotenuse and vertical side of the right-angled triangle shown below.



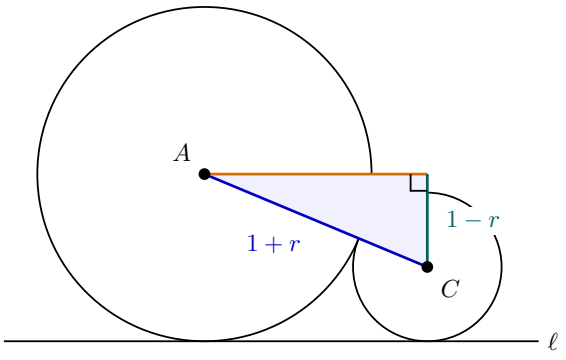
By Pythagoras, the horizontal side has length

$$\sqrt{5^2 - 3^2} = \sqrt{25 - 9} = \sqrt{16} = 4$$

Now consider circles A and C. Their centres are $1 + r$ apart.

Circle C is the smaller circle in the gap, so $r < 1$ and the difference in the heights of their centres is $1 - r$.

The next sketch shows the corresponding right-angled triangle, with circle B omitted.



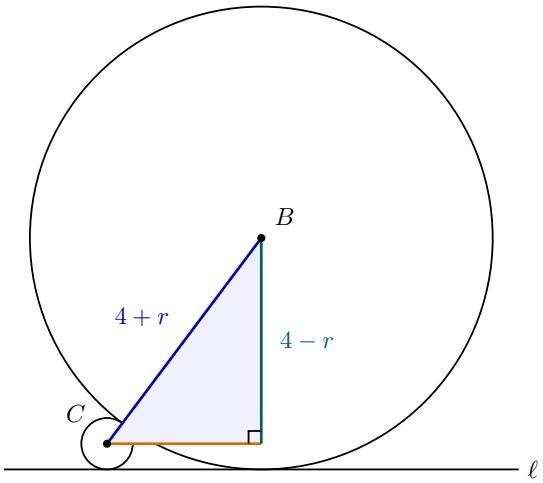
By Pythagoras, its horizontal side has length

$$\begin{aligned}\sqrt{(1+r)^2 - (1-r)^2} &= \sqrt{(1 + 2r + r^2) - (1 - 2r + r^2)} \\ &= \sqrt{4r} \\ &= 2\sqrt{r}\end{aligned}$$

Finally, consider circles B and C.

Their centres are $4 + r$ apart, and the difference in their heights is $4 - r$.

This gives the third right-angled triangle, shown below with circle A omitted.



By Pythagoras, its horizontal side has length

$$\sqrt{(4+r)^2-(4-r)^2} = \sqrt{(16+8r+r^2)-(16-8r+r^2)}$$

$$= \sqrt{16r}$$

$$= 4\sqrt{r}$$

Since the centre of C lies horizontally between the centres of A and B, the two shorter horizontal lengths add to 4:

$$2\sqrt{r} + 4\sqrt{r} = 4$$

Collecting terms,

$$6\sqrt{r} = 4$$

Dividing by 6,

$$\sqrt{r} = \frac{2}{3}$$

Squaring,

$$r = \left(\frac{2}{3}\right)^2 = \frac{4}{9}$$

The correct option is **B**.

Probability & Counting

8. $N = 2^7 \times 3^5$

One of the positive factors of N that is divisible by 12 is selected at random, with each such factor equally likely to be selected.

What is the probability that N divided by the selected factor is a perfect square?

A $\frac{1}{6}$

B $\frac{1}{5}$

C $\frac{1}{4}$

D $\frac{3}{10}$

E $\frac{1}{3}$

F $\frac{2}{5}$

Solution **Answer D**

Since $12 = 2^2 \times 3$, a selected factor must contain at least two powers of 2 and one power of 3. Write it as

$$d = 2^{2+i}3^{1+j}$$

where the exponents cannot exceed those in N, so

$$0 \leq i \leq 5, \qquad 0 \leq j \leq 4$$

There are 6 choices for i and 5 for j, giving

$$6 \times 5 = 30$$

equally likely factors.

For any such factor,

$$\frac{N}{d} = \frac{2^7 3^5}{2^{2+i}3^{1+j}}$$

Cancelling the powers gives

$$\frac{N}{d} = 2^{5-i}3^{4-j}$$

A positive integer is a perfect square if every exponent in its prime factorisation is even, including an exponent of zero. Thus $5 - i$ must be even, giving

$$i = 1,\ 3 \text{ or } 5$$

Also, $4 - j$ must be even, giving

$$j = 0,\ 2 \text{ or } 4$$

There are $3 \times 3 = 9$ factors for which N/d is a perfect square. The probability is therefore

$$\frac{9}{30} = \frac{3}{10}$$

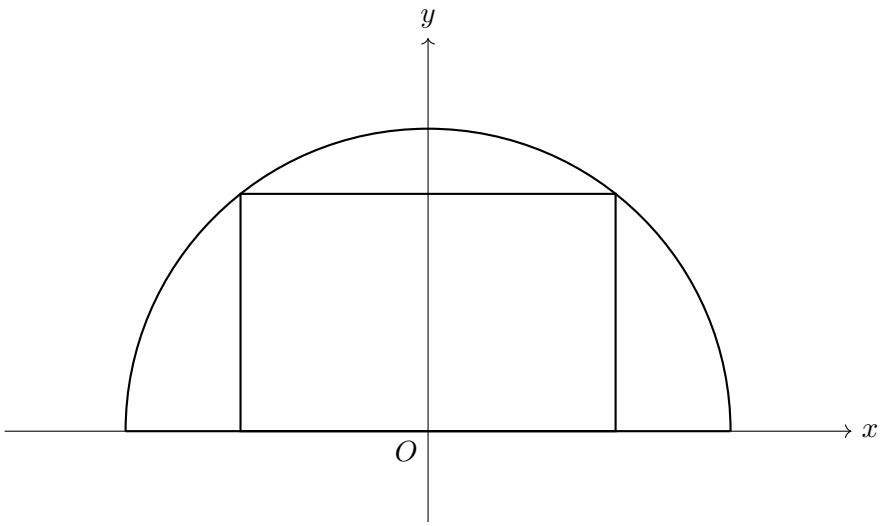
The correct option is **D**.

Geometry

9. The diagram shows a semicircle of radius 1 with its diameter along the x-axis and its centre at the origin.

A rectangle has one side on the diameter and its two remaining vertices on the arc of the semicircle.

What is the largest possible perimeter of such a rectangle?



A $\sqrt{5}$

B 4

C $2+\sqrt{5}$

D $3\sqrt{2}$

E $2\sqrt{5}$

F 5

Solution **Answer E**

Let x be half the rectangle's width and y its height. The upper corners have the same height and lie on the circle, so their horizontal coordinates are $-x$ and x. Hence

$$x^2+y^2=1$$

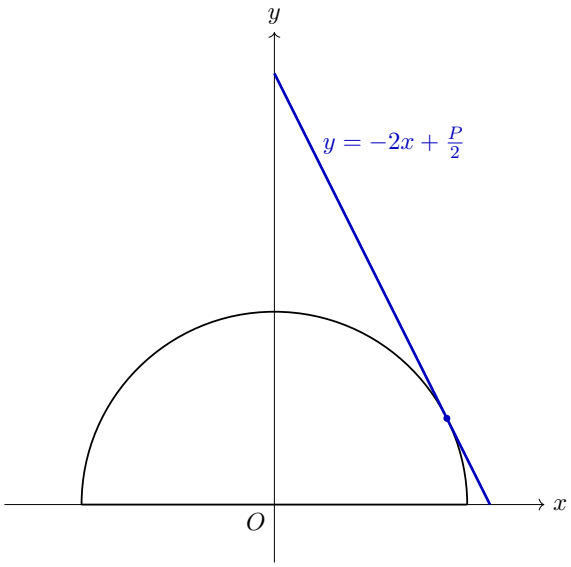
The perimeter is

$$P=4x+2y$$

Making y the subject gives

$$y=-2x+\frac{P}{2}$$

For each value of P, this is a line of gradient -2. Increasing P moves the line upwards. The sketch shows its limiting position: the highest line that still meets the relevant arc is tangent to it.



Substituting the line equation into the circle equation gives

$$x^2 + \left(-2x + \frac{P}{2}\right)^2 = 1$$

Expanding,

$$x^2 + 4x^2 - 2Px + \frac{P^2}{4} = 1$$

Rearranging,

$$5x^2 - 2Px + \frac{P^2}{4} - 1 = 0$$

At tangency, this quadratic has a repeated real root, so its discriminant is zero. The discriminant is

$$(-2P)^2 - 4(5)\left(\frac{P^2}{4} - 1\right) = 20 - P^2$$

Therefore

$$20 - P^2 = 0$$

Since a rectangle has positive perimeter,

$$P = 2\sqrt{5}$$

To check that the tangent point gives a rectangle, the repeated root is

$$x = \frac{2P}{2(5)} = \frac{2}{\sqrt{5}}$$

Substituting into the line equation gives

$$y = -2\left(\frac{2}{\sqrt{5}}\right) + \sqrt{5} = \frac{1}{\sqrt{5}}$$

Both dimensions are positive, so this perimeter is attainable. The correct option is **E**.

Alternative Solution 1: (Chain Rule)

Eliminate y using the positive square root, since the rectangle's height is non-negative:

$$P = 4x + 2\sqrt{1 - x^2}, \qquad 0 \le x \le 1$$

The chain rule is beyond the TMUA specification, but is useful for differentiating an expression like this. To differentiate the square-root term, put

$$u = 1 - x^2 \qquad \text{and} \qquad Q = \sqrt{u} = u^{\frac{1}{2}}$$

Differentiating each expression gives

$$\frac{\mathrm{d}Q}{\mathrm{d}u} = \frac{1}{2}u^{-1/2} = \frac{1}{2\sqrt{u}}$$

and

$$\frac{\mathrm{d}u}{\mathrm{d}x} = -2x$$

The chain rule states that

$$\frac{\mathrm{d}Q}{\mathrm{d}x} = \frac{\mathrm{d}Q}{\mathrm{d}u}\,\frac{\mathrm{d}u}{\mathrm{d}x}$$

Therefore

$$\frac{\mathrm{d}Q}{\mathrm{d}x} = \frac{1}{2\sqrt{u}}(-2x) = -\frac{x}{\sqrt{1 - x^2}}$$

Thus, for $0 < x < 1$,

$$\frac{\mathrm{d}P}{\mathrm{d}x} = 4 - \frac{2x}{\sqrt{1 - x^2}}$$

Setting $\frac{\mathrm{d}P}{\mathrm{d}x} = 0$ gives

$$4 = \frac{2x}{\sqrt{1 - x^2}}$$

Rearranging,

$$2\sqrt{1 - x^2} = x$$

Squaring,

$$4(1 - x^2) = x^2 \implies 5x^2 = 4$$

Because $x > 0$, the stationary value occurs at

$$x = \frac{2}{\sqrt{5}}$$

Substituting this into the perimeter gives

$$\begin{aligned} P\left(\frac{2}{\sqrt{5}}\right) &= 4\left(\frac{2}{\sqrt{5}}\right) + 2\sqrt{1 - \left(\frac{2}{\sqrt{5}}\right)^2} \\ &= \frac{8}{\sqrt{5}} + \frac{2}{\sqrt{5}} \\ &= 2\sqrt{5} \end{aligned}$$

The correct option is again found to be **E**.

Alternative Solution 2: (Cauchy-Schwarz)

The Cauchy–Schwarz inequality, which is beyond the specification, states that for any real numbers a, b, x, y

$$(ax + by)^2 \leq \left(a^2 + b^2\right)\left(x^2 + y^2\right)$$

Using $a = 4$, $b = 2$ and $x^2 + y^2 = 1$ gives

$$P^2 = (4x + 2y)^2 \leq (4^2 + 2^2)(1) = 20$$

As P is positive,

$$P \leq 2\sqrt{5}$$

Equality is possible when (x, y) is proportional to $(4, 2)$. Take $x = 2/\sqrt{5}$ and $y = 1/\sqrt{5}$: these are positive and satisfy

$$x^2 + y^2 = \frac{4}{5} + \frac{1}{5} = 1$$

Their perimeter is

$$P = 4\left(\frac{2}{\sqrt{5}}\right) + 2\left(\frac{1}{\sqrt{5}}\right) = 2\sqrt{5}$$

This is the shortest route if you know the inequality. The correct option is **E**.

Polynomials

10. Find the complete set of real values of k for which the quadratic equation

$$x^2 - (k+2)x + (2k+1) = 0$$

has two distinct **positive** real roots.

A $k > 4$

B $k < 0$ or $k > 4$

C $-\frac{1}{2} < k < 0$ or $k > 4$

D $-2 < k < 0$ or $k > 4$

E $k > -\frac{1}{2}$

Solution **Answer C**

For two distinct real roots, the discriminant must be positive. Once there are two real roots, a positive product means they have the same sign, and a positive sum means they are both positive.

For $ax^2 + bx + c = 0$, the sum of the roots is $-b/a$. Here it is

$$-\frac{-(k+2)}{1} = k + 2$$

So the sum is positive when

$$k + 2 > 0$$

Rearranging gives

$$k > -2$$

The product of the roots is c/a. Here it is

$$\frac{2k+1}{1} = 2k + 1$$

So the product is positive when

$$2k + 1 > 0$$

Rearranging gives

$$k > -\frac{1}{2}$$

This condition also ensures $k > -2$, so both sign conditions hold when $k > -\frac{1}{2}$.

Now require the roots to be real and distinct. The discriminant is

$$\begin{aligned}(k+2)^2 - 4(2k+1) &= k^2 + 4k + 4 - 8k - 4 \\ &= k^2 - 4k \\ &= k(k-4)\end{aligned}$$

It is positive when

$$k < 0 \quad \text{or} \quad k > 4$$

Combining this with $k > -\frac{1}{2}$, the complete range is

$$-\frac{1}{2} < k < 0 \quad \text{or} \quad k > 4$$

The correct option is **C**.

Alternative Solution:

The quadratic formula is

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Substituting the coefficients gives

$$x = \frac{k + 2 \pm \sqrt{(k+2)^2 - 4(2k+1)}}{2}$$

Simplifying the discriminant,

$$\begin{aligned}(k+2)^2 - 4(2k+1) &= k^2 - 4k \\ &= k(k-4)\end{aligned}$$

Hence the roots are

$$x = \frac{k + 2 \pm \sqrt{k(k-4)}}{2}$$

For two distinct real roots, we first need

$$k(k-4) > 0$$

This gives

$$k < 0 \quad \text{or} \quad k > 4$$

Of the two roots, the one with the minus sign is smaller. Both roots are positive exactly when this smaller root is positive:

$$\frac{k + 2 - \sqrt{k(k-4)}}{2} > 0$$

Multiplying by 2 gives

$$k + 2 > \sqrt{k(k-4)}$$

The square root is non-negative, so this requires

$$k + 2 > 0$$

With both sides non-negative, squaring the inequality is reversible. It gives

$$(k+2)^2 > k(k-4)$$

Expanding,

$$k^2 + 4k + 4 > k^2 - 4k$$

Rearranging,

$$8k + 4 > 0$$

Therefore

$$k > -\frac{1}{2}$$

This also guarantees $k + 2 > 0$, as required before squaring. Combining it with the condition for two distinct real roots gives

$$-\frac{1}{2} < k < 0 \quad \text{or} \quad k > 4$$

The correct option is **C**.

Sequences & Series

11. A sequence a_n is defined by $a_1 = 1$ and, for $n \geq 1$,

$$a_{n+1} = a_n + \frac{1}{a_n}$$

Which of the following statements is true for all $n \geq 1$?

A a_n is an integer

B a_n^2 is an integer

C $a_n < n$

D $a_n^2 \geq 2n - 1$

E $a_n^2 \leq 2n - 1$

F $a_n \leq \sqrt{2n}$

Solution **Answer D**

Start with the first few terms. These are enough to rule out five of the six statements:

$$a_1 = 1$$

$$a_2 = 1 + \frac{1}{1} = 2$$

$$a_3 = 2 + \frac{1}{2} = \frac{5}{2}$$

Statement C fails at $n = 1$, since $a_1 = 1$ is not less than 1. Statement E fails at $n = 2$, since

$$a_2^2 = 2^2 = 4 > 3 = 2(2) - 1$$

At $n = 3$, $a_3 = \frac{5}{2}$ is not an integer, ruling out A. Also,

$$a_3^2 = \left(\frac{5}{2}\right)^2 = \frac{25}{4}$$

so B fails. Finally,

$$\frac{25}{4} > 6 = 2(3)$$

Since $a_3 > 0$, this means $a_3 > \sqrt{2(3)}$, ruling out F.

The correct option must therefore be **D**.

Remark:

We can also prove directly by mathematical induction that

$$a_n^2 \geq 2n - 1$$

for every integer $n \geq 1$.

First note that all terms are positive: $a_1 = 1 > 0$, and whenever $a_n > 0$, the recurrence gives $a_{n+1} = a_n + 1/a_n > 0$.

For $n = 1$,

$$a_1^2 = 1 = 2(1) - 1$$

so the bound holds.

Assume that the bound holds for $n = k$, where $k \geq 1$ is an integer. Thus

$$a_k^2 \geq 2k - 1$$

Using the recurrence,

$$\begin{aligned} a_{k+1}^2 &= \left(a_k + \frac{1}{a_k}\right)^2 \\ &= a_k^2 + 2 + \frac{1}{a_k^2} \\ &\geq a_k^2 + 2 \\ &\geq (2k - 1) + 2 \\ &= 2(k + 1) - 1 \end{aligned}$$

where we used $1/a_k^2 > 0$ and then the induction hypothesis.

Hence, if the bound holds for $n = k$, it also holds for $n = k + 1$. Since it holds for $n = 1$, it follows by mathematical induction that

$$a_n^2 \geq 2n - 1$$

for every integer $n \geq 1$.

Proof by induction is not on the TMUA spec, but will be familiar to some of the students preparing for it.

Number

12. The positive real numbers $p \times 10^{-4}$, $q \times 10^{-2}$ and $r \times 10^{-1}$ are each in standard form, and

$$(p \times 10^{-4}) + (q \times 10^{-2}) = (r \times 10^{-1})$$

How many of the following statements must be true?

I: $q > 1$
II: $r > 1$
III: $q < r$
IV: $p < r$

A 1 of them

B 2 of them

C 3 of them

D 4 of them

E None of them

Solution **Answer A**

The p term is very small, so q must be close to 10 and r close to 1; this suggests that I must be true and III is false, while equality at 1 may provide counterexamples to II and IV.

Since the three numbers are in standard form,

$$1 \leq p < 10, \qquad 1 \leq q < 10, \qquad 1 \leq r < 10$$

Multiplying the given equation by 10^4 gives

$$p + 100q = 1000r$$

To test statement I, rearrange:

$$q = 10r - \frac{p}{100}$$

Since $r \geq 1$ and $p < 10$,

$$q > 10 - \frac{10}{100} = 9.9$$

In particular, $q > 1$, so statement I must be true.

For the remaining statements, a single example can show that none must be true. Take

$$p = 1, \qquad q = 9.99, \qquad r = 1$$

These coefficients are all at least 1 and less than 10, as required for standard form. They also satisfy the equation, since

$$1 \times 10^{-4} + 9.99 \times 10^{-2} = 0.0001 + 0.0999 = 0.1 = 1 \times 10^{-1}$$

For this example:

II:	$r > 1$ becomes $1 > 1$,	which is false;
III:	$q < r$ becomes $9.99 < 1$,	which is false;
IV:	$p < r$ becomes $1 < 1$,	which is false.

Thus only statement I must be true. The correct option is **A**.

Probability & Counting

13. A bag contains only red balls and blue balls, with at least one ball of each colour. The total number of balls is n, where $30 < n < 50$.

Two balls are drawn at random without replacement. The probability that they have the same colour is equal to the probability that they have different colours.

How many different possible values are there for the number of red balls in the bag?

A 1

B 2

C 3

D 4

E 5

Solution **Answer C**

Let r be the number of red balls, so there are $n - r$ blue balls. Because the balls are drawn without replacement, there is one fewer ball available on the second draw.

The draws have the same colour if they are both red or both blue. Adding these probabilities gives

$$P(\text{same}) = \frac{r}{n}\frac{r-1}{n-1} + \frac{n-r}{n}\frac{n-r-1}{n-1}$$

Hence

$$P(\text{same}) = \frac{r(r-1) + (n-r)(n-r-1)}{n(n-1)}$$

The draws have different colours if red is followed by blue or blue is followed by red. These two orders each have probability $\dfrac{r(n-r)}{n(n-1)}$, which accounts for the factor of 2:

$$P(\text{different}) = \frac{2r(n-r)}{n(n-1)}$$

Equating the probabilities and cancelling the non-zero denominator gives

$$r(r-1) + (n-r)(n-r-1) = 2r(n-r)$$

Expanding,

$$r^2 - r + (n-r)^2 - (n-r) = 2r(n-r)$$

Collecting the terms involving r,

$$r^2 + (n-r)^2 - n = 2r(n-r)$$

Rearranging and factorising,

$$\big(r - (n-r)\big)^2 = n$$

Therefore

$$(2r-n)^2 = n$$

Both r and n are integers, so n must be a square number. The square numbers satisfying $30 < n < 50$ are

$$n = 36 \quad \text{or} \quad n = 49$$

For $n = 36$,

$$(2r - 36)^2 = 36$$

Taking either square root gives

$$2r - 36 = 6 \quad \text{or} \quad 2r - 36 = -6$$

Thus

$$r = 21 \quad \text{or} \quad r = 15$$

For $n = 49$,

$$(2r - 49)^2 = 49$$

Taking either square root gives

$$2r - 49 = 7 \quad \text{or} \quad 2r - 49 = -7$$

Thus

$$r = 28 \quad \text{or} \quad r = 21$$

The distinct possible numbers of red balls are 15, 21 and 28. There are three, so the correct option is **C**.

Functions & Graphs

14. For each real value of x, let $M(x)$ be the larger of the two numbers

$$(x+3)^2+1 \quad \text{and} \quad 4(x-1)^2+1$$

For which value of x is $M(x)$ least?

A -3

B -1

C $-\frac{1}{3}$

D $\frac{1}{3}$

E 1

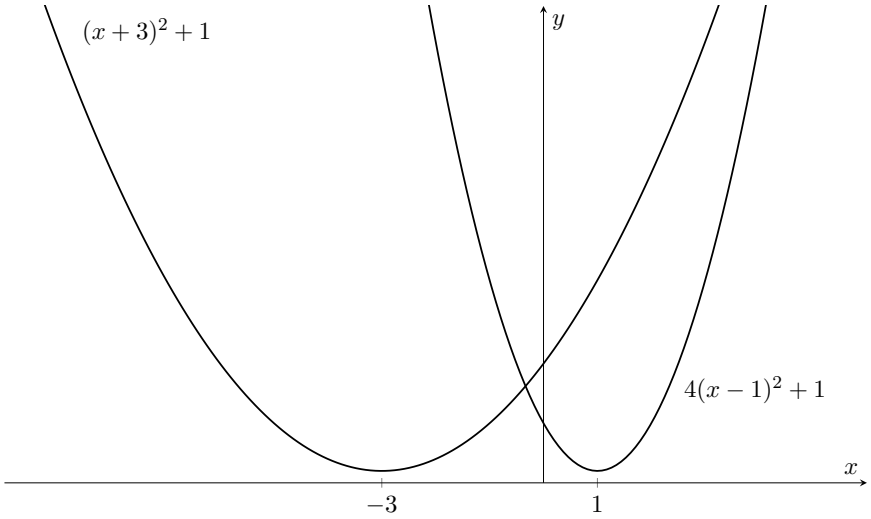
F 5

Solution **Answer C**

The two parabolas have vertices at $x=-3$ and $x=1$.

Since $M(x)$ takes the larger value, the vertex of either parabola need not give the minimum of $M(x)$.

Sketch both curves to see which one is higher.

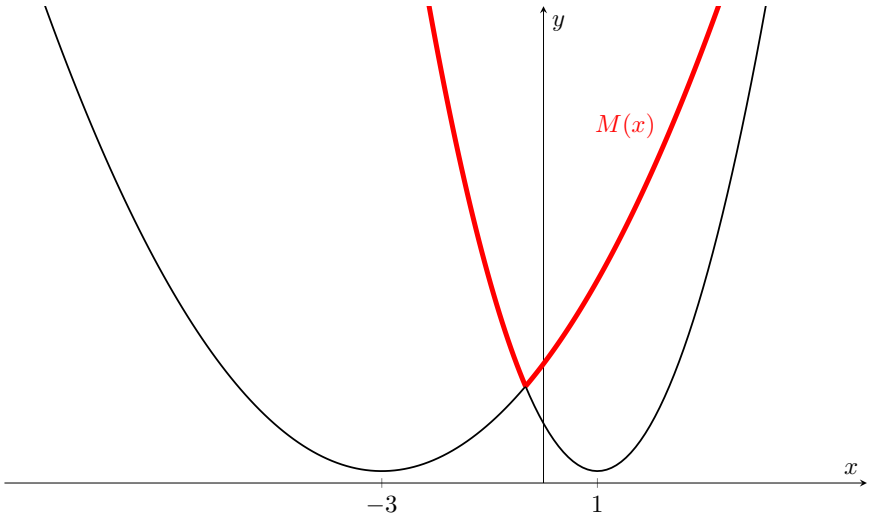


For any fixed value of x, compare the heights of the two curves: $M(x)$ is the height of the upper one.

For example, at $x=0$ the two values are 10 and 5, so $M(0)=10$.

To draw $M(x)$, follow whichever curve is higher, switching from one parabola to the other where they cross.

The selected pieces are highlighted in red below; the lower pieces remain black and do not form part of $M(x)$.



From the graph, $M(x)$ is least at the intersection shown. We now find its exact position.

At an intersection, the two expressions are equal:

$$(x+3)^2+1 = 4(x-1)^2+1$$

Cancelling 1 gives

$$(x+3)^2 = 4(x-1)^2$$

Expanding,

$$x^2+6x+9 = 4x^2-8x+4$$

Rearranging,

$$3x^2-14x-5 = 0$$

Factorising,

$$(3x+1)(x-5) = 0$$

Hence the intersections occur at

$$x = -\frac{1}{3} \quad \text{or} \quad x = 5$$

The intersection shown is the one at $x = -\frac{1}{3}$.

Thus $M(x)$ is least when $x = -\frac{1}{3}$, so the correct option is **C**.

Number

15. Let $N = 2^6 \times 3^4 \times 5^3$

A positive factor d of N is chosen such that the highest common factor of d and $\frac{N}{d}$ is 60

How many possible values of d are there?

A 2

B 4

C 6

D 8

E 12

Solution **Answer D**

Write the factor as

$$d = 2^i 3^j 5^k$$

where $0 \leq i \leq 6$, $0 \leq j \leq 4$ and $0 \leq k \leq 3$. The complementary factor is

$$\frac{N}{d} = 2^{6-i} 3^{4-j} 5^{3-k}$$

The highest common factor takes the smaller exponent of each prime in these two factors. Since

$$60 = 2^2 \times 3 \times 5$$

matching exponents gives

$$\min(i, 6-i) = 2$$

$$\min(j, 4-j) = 1$$

$$\min(k, 3-k) = 1$$

For the power of 2, either $i = 2$ or $6 - i = 2$. Both give the required minimum, so

$$i = 2 \text{ or } i = 4$$

Similarly, the power of 3 gives

$$j = 1 \text{ or } j = 3$$

and the power of 5 gives

$$k = 1 \text{ or } k = 2$$

Each exponent therefore has two choices, and the choices are independent. The number of possible factors is

$$2 \times 2 \times 2 = 8$$

The correct option is **D**.

Polynomials

16. Consider

$$P(x) = x^4 + ax^3 + bx^2 + ax + 1$$

where a and b are real constants.

Suppose $(x-2)$ and $\left(x+\frac{1}{2}\right)$ are both factors of $P(x)$

What is the value of b?

A -13

B -5

C 0

D 5

E 13

F $-\frac{17}{4}$

Solution **Answer F**

By the factor theorem, if $(x-r)$ is a factor of $P(x)$, then $P(r) = 0$.

The two given factors therefore give $P(2) = 0$ and $P\left(-\frac{1}{2}\right) = 0$.

Substituting $x = 2$ gives

$$16 + 8a + 4b + 2a + 1 = 0$$

Collecting terms,

$$17 + 10a + 4b = 0$$

Now substitute $x = -\frac{1}{2}$:

$$\begin{aligned} P\left(-\frac{1}{2}\right) &= \frac{1}{16} - \frac{a}{8} + \frac{b}{4} - \frac{a}{2} + 1 \\ &= \frac{17}{16} - \frac{a}{8} + \frac{b}{4} - \frac{a}{2} \\ &= \frac{17}{16} - \frac{5a}{8} + \frac{b}{4} \end{aligned}$$

Since this is zero, multiplying by 16 gives

$$17 - 10a + 4b = 0$$

Adding the two equations eliminates a:

$$34 + 8b = 0$$

Rearranging,

$$8b = -34$$

Hence

$$b = -\frac{17}{4}$$

The correct option is **F**.

Integration

17. Let k be a positive constant. The curve C has equation

$$y = x^3 - 3k^2x + 2k^3$$

The point A is the local maximum of C

The line parallel to the x-axis passing through A intersects C again at the point B

The area of the region bounded by C and the line segment AB is 108

What is the value of k?

A 1

B $\sqrt{2}$

C $\sqrt{3}$

D 2

E $2\sqrt{2}$

F 3

G 4

Solution **Answer D**

Differentiate to find the stationary points:

$$\frac{\mathrm{d}y}{\mathrm{d}x} = 3x^2 - 3k^2$$

Factorising gives

$$\frac{\mathrm{d}y}{\mathrm{d}x} = 3(x-k)(x+k)$$

Hence

$$x = -k \quad \text{or} \quad x = k$$

Substituting $x = -k$ into the equation of C gives

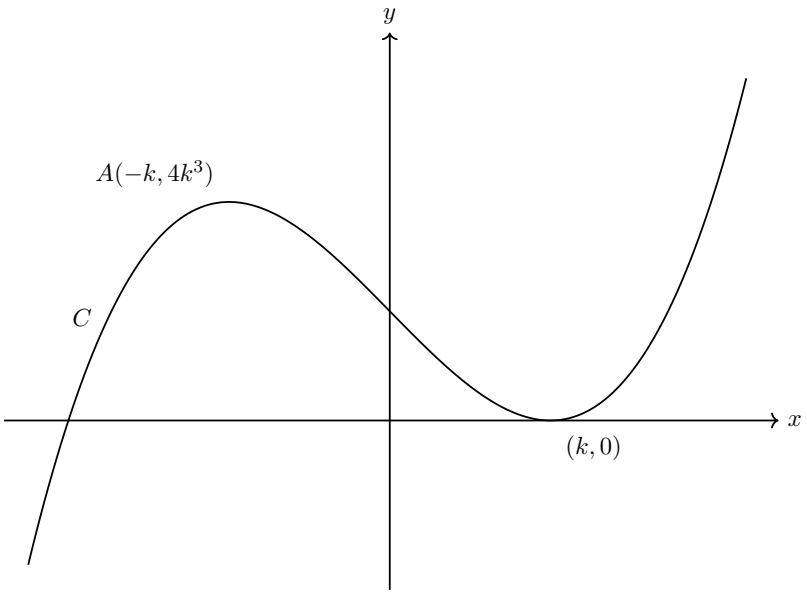
$$y = (-k)^3 - 3k^2(-k) + 2k^3 = 4k^3$$

Substituting $x = k$ gives

$$y = k^3 - 3k^2(k) + 2k^3 = 0$$

Since $k > 0$, the gradient changes from positive to negative at $x = -k$, then from negative to positive at $x = k$.

Thus A is $(-k, 4k^3)$ and the local minimum is $(k, 0)$, giving this sketch of C:



The line through A parallel to the x-axis is

$$y = 4k^3$$

To find its other intersection with C, set the two expressions for y equal:

$$x^3 - 3k^2x + 2k^3 = 4k^3$$

Rearranging gives

$$x^3 - 3k^2x - 2k^3 = 0$$

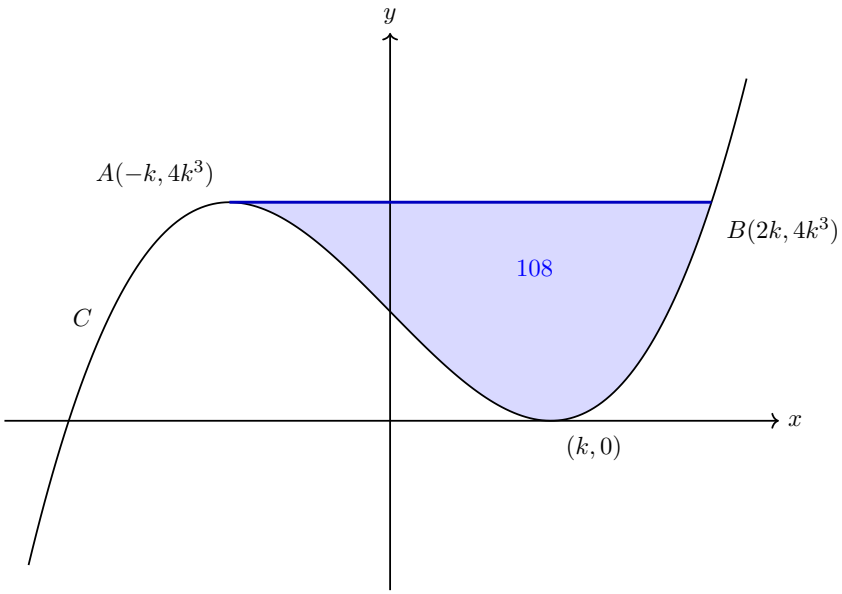
The line touches C at the stationary point A, so $x = -k$ is a double root of this equation. Factorising gives

$$(x + k)^2(x - 2k) = 0$$

The other intersection therefore has $x = 2k$, so

$$B = (2k, 4k^3)$$

Adding AB to the sketch shows the bounded region:



Between $x = -k$ and $x = 2k$, the line is above the curve: their difference is $(x+k)^2(2k-x)$, which is non-negative there.

For an upper graph $f(x)$ and a lower graph $g(x)$, the formula for the area between the graphs is

$$\text{Area} = \int_a^b \big(f(x) - g(x)\big)\,\mathrm{d}x$$

Here $f(x) = 4k^3$ and $g(x) = x^3 - 3k^2x + 2k^3$. Using the given area,

$$108 = \int_{-k}^{2k} \big(4k^3 - (x^3 - 3k^2x + 2k^3)\big)\,\mathrm{d}x$$

Simplifying the integrand gives

$$108 = \int_{-k}^{2k} (-x^3 + 3k^2x + 2k^3)\,\mathrm{d}x$$

Integrating gives

$$108 = \left[-\frac{x^4}{4} + \frac{3k^2x^2}{2} + 2k^3x\right]_{-k}^{2k}$$

Substituting the limits gives

$$108 = \left(-\frac{(2k)^4}{4} + \frac{3k^2(2k)^2}{2} + 2k^3(2k)\right) - \left(-\frac{(-k)^4}{4} + \frac{3k^2(-k)^2}{2} + 2k^3(-k)\right)$$

Simplifying gives

$$108 = 6k^4 - \left(-\frac{3k^4}{4}\right)$$

Therefore

$$108 = \frac{27k^4}{4}$$

Rearranging gives

$$k^4 = 16$$

Since k is positive,

$$k = 2$$

The correct option is **D**.

Algebra

18. Let a and b be real numbers with $a + b = 1$

What is the least possible value of

$$(a^2 + b)(b^2 + a) \quad ?$$

A $\frac{9}{16}$

B $\frac{1}{4}$

C $\frac{1}{2}$

D $\frac{3}{4}$

E 1

Solution **Answer A**

Using $b = 1 - a$, substitute into the product:

$$(a^2 + b)(b^2 + a) = \big(a^2 + 1 - a\big)\big((1 - a)^2 + a\big)$$

Expanding the second factor gives

$$(1 - a)^2 + a = 1 - 2a + a^2 + a = a^2 - a + 1$$

The two factors are therefore equal, so

$$(a^2 + b)(b^2 + a) = (a^2 - a + 1)^2$$

Complete the square inside the brackets:

$$a^2 - a + 1 = \left(a - \frac{1}{2}\right)^2 - \frac{1}{4} + 1$$

$$= \left(a - \frac{1}{2}\right)^2 + \frac{3}{4}$$

The squared term is at least 0, with equality when $a = \frac{1}{2}$. Thus the expression inside the outer square has minimum $\frac{3}{4}$, which is positive.

Squaring a positive number increases as that number increases, so the product has minimum

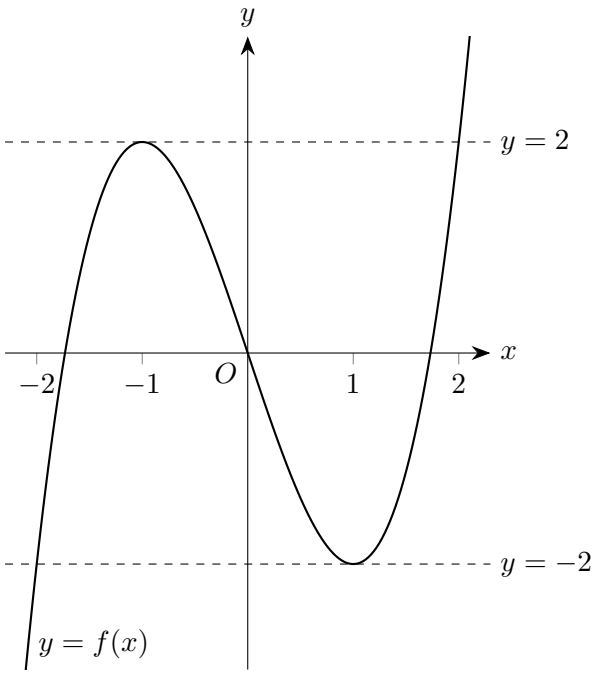
$$\left(\frac{3}{4}\right)^2 = \frac{9}{16}$$

This is attained when $a = \frac{1}{2}$ and $b = \frac{1}{2}$. The correct option is **A**.

Functions & Graphs

19. The diagram shows the graph of $y = f(x)$, where $f(x) = x^3 - 3x$. The horizontal lines through the turning points are also shown.

How many distinct real solutions does the equation $f(f(x)) = f(x)$ have?



A 5

B 6

C 7

D 8

E 9

Solution **Answer C**

Put $u = f(x)$. Then the equation becomes $f(u) = u$: we first need to find which outputs of f are equal to their own input.

On the graph, $f(u) = u$ holds where $y = f(x)$ meets $y = x$. Adding that line shows crossings at $(-2, -2)$, $(0, 0)$ and $(2, 2)$.

To confirm that these are all the possible values of u, factor the equation:

$$u^3 - 3u = u$$

Rearranging,

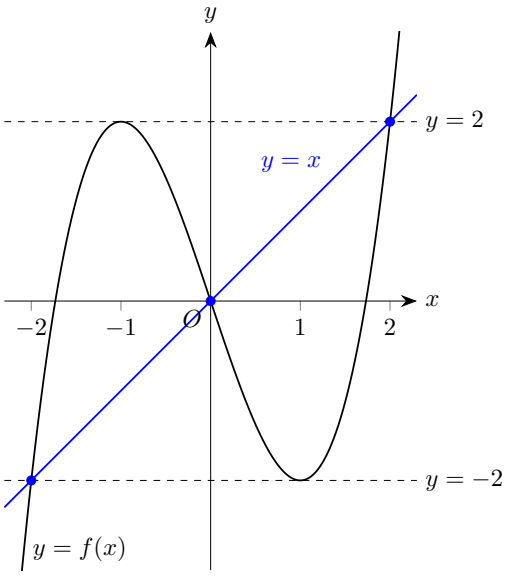
$$u^3 - 4u = 0$$

Factorising,

$$u(u - 2)(u + 2) = 0$$

Hence

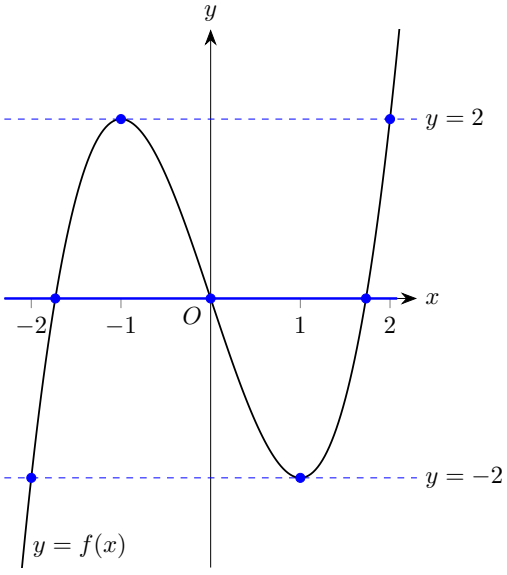
$$u = -2 \text{ or } u = 0 \text{ or } u = 2$$



Now count the inputs x that produce each of these three outputs.

For each value of u, draw the horizontal line $y = u$; its intersections with the curve are the solutions of $f(x) = u$.

The lines $y = -2$ and $y = 2$ are already on the given graph, while $y = 0$ is the x-axis.



The line $y = -2$ crosses the curve at $x = -2$ and touches it at $x = 1$. It therefore gives

$$u = -2 : \quad 2 \text{ distinct solutions}$$

The line $y = 0$ crosses the curve three times: once on each side of the origin and once at the origin. It gives

$$u = 0 : \quad 3 \text{ distinct solutions}$$

The line $y = 2$ touches the curve at $x = -1$ and crosses it at $x = 2$. It gives

$$u = 2 : \quad 2 \text{ distinct solutions}$$

No input x can produce two different values of u, so these counts do not overlap. The total is

$$2 + 3 + 2 = 7$$

The correct option is **C**.

The substitution $u = f(x)$ is useful because $f(x)$ is both the input of the outer f and the whole right-hand side. Solving $f(u) = u$ identifies the possible outputs first; the horizontal lines then count the inputs producing each output.

Inequalities

20. Let $f(x) = ax^2 + bx + c$ be a quadratic function with real coefficients such that $f(x) \geq 0$ for all real x

Suppose that for every real x,

$$f(x+1) - f(x) \geq 2x + 3$$

What is the smallest possible value of $f(0)$?

A 0

B $\frac{1}{4}$

C 1

D 2

E $\frac{9}{4}$

F 4

G $\frac{25}{4}$

Solution **Answer C**

Since f is a quadratic that is non-negative for every real x, it opens upwards, so $a > 0$. Its discriminant must satisfy

$$b^2 - 4ac \leq 0$$

Now expand the difference in the given inequality:

$$\begin{aligned} f(x+1) - f(x) &= a(x+1)^2 + b(x+1) + c - (ax^2 + bx + c) \\ &= 2ax + a + b \end{aligned}$$

Thus the inequality becomes

$$2ax + a + b \geq 2x + 3$$

Rearranging,

$$2(a-1)x + (a + b - 3) \geq 0$$

This must hold for every real x. The left-hand side is a linear expression, whose graph is a straight line. Any straight line with a non-zero gradient goes below the horizontal axis in one direction:

- If $a > 1$, the gradient is positive, so sufficiently large negative values of x make the expression negative.
- If $a < 1$, the gradient is negative, so sufficiently large positive values of x make the expression negative.

Both possibilities contradict the inequality. The line must therefore be horizontal, so its gradient is zero:

$$2(a-1) = 0$$

Hence

$$a = 1$$

Substituting this into the inequality leaves

$$b - 2 \geq 0$$

so

$$b \geq 2$$

The discriminant condition for f now becomes

$$b^2 - 4c \leq 0$$

Rearranging and using $b \geq 2$,

$$c \geq \frac{b^2}{4} \geq 1$$

To check that this lower bound can be achieved, take $a = 1$, $b = 2$ and $c = 1$. Then

$$f(x) = (x + 1)^2 \geq 0$$

for every real x, and

$$f(x + 1) - f(x) = (x + 2)^2 - (x + 1)^2 = 2x + 3$$

Thus both conditions are satisfied, and the smallest possible value of $f(0) = c$ is

$$f(0) = 1$$

The correct option is **C**.